CHNT31 · Digital Creative Award · Poster

From Pocket to SPARQL: How a Smartphone Scan of an Ogham Stone Learned to Speak Linked Data

FDOx: FAIR Digital Objects for 3D cultural heritage

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A scan in the pocket

Citizen-generated 3D models of heritage monuments appear daily on platforms such as Sketchfab. Their metadata are thin, platform-bound and rarely machine-readable, so the models are hard to cite, preserve or link with research data.



Figure 1. CIIC 81, an Ogham stone in Cork: preview of the packaged 3D model.

The idea: FDOx

FDOx is a practice-driven, Linked-Data-based implementation of the FAIR Digital Object idea (De Smedt et al., 2020) for 3D cultural heritage: one model, one self-describing package (Thiery, 2026). All components are open source and deterministic, so anyone can repeat the workflow for their own models.

Our own perspective, not an official FDO reference implementation.

Explore

FDOx registry

Open-source code

Ogham stone on Zenodo

The life of one model: from the pocket to the knowledge graph

1. **Capture** — smartphone scan on Sketchfab
2. **Package** — Python pipeline, open formats, viewer
3. **Describe** — two files become RDF, SHACL-checked
4. **Deposit** — Zenodo with a persistent identifier
5. **Register** — harvest, DCAT catalogue, CIDOC CRM
6. **Query** — federated knowledge graph, SPARQL

2 · Package

Fetch the model and its metadata from Sketchfab, convert it to open formats, build a web viewer and assemble a self-describing package.

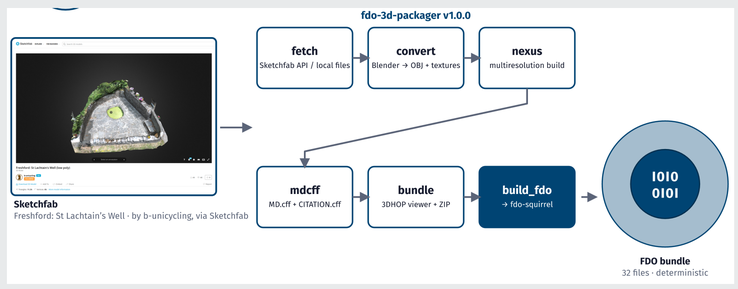


Figure 2. fdo-3d-packager v1.0.0, here with the Freshford holy well (model: b-unicycling, CC BY 4.0, via Sketchfab).

3 · Describe

Two human-editable files, `MD.cff` and `CITATION.cff`, become RDF with DCAT, PROV-O and CIDOC CRM, with provenance for every step; SHACL validates the package.

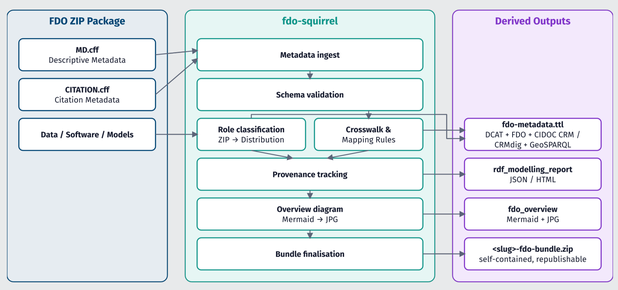


Figure 3. fdo-squirrel: from package to RDF and bundle.

4–6 · Register and query

Deposited on Zenodo with a persistent identifier, harvested by a registry, anchored in CIDOC CRM and published as a DCAT catalogue that feeds a federated knowledge graph.

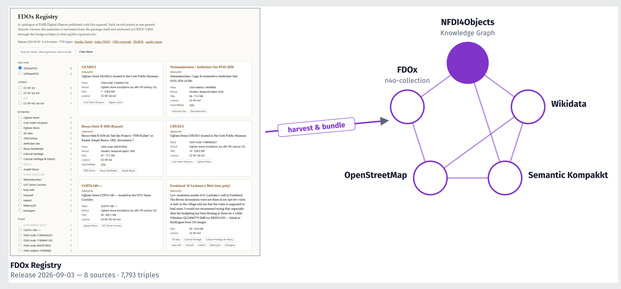


Figure 4. The FDOx registry (release 2026-09-03) and its hubs.

Query 1: licences (prefixes omitted)

```
SELECT ?title
(REPLACE(STR(?iri), "^.*/|\\.html$", "") AS ?licence)
WHERE { ?fdo a fdo:3DDataFDO ;
dct:title ?title ; dct:license ?iri .
FILTER(STRSTARTS(STR(?iri), "https://spdx.org/licenses/")) }
ORDER BY ?title
```

Model	Licence
Beuys-Stele B 1036 (Kassel)	CC-BY-4.0
CHUIS/1	CC-BY-SA-4.0
CO074-148---- (CIIC 81)	CC-BY-NC-SA-4.0
Freshford: St Lachtain's Well (low poly)	CC-BY-4.0
GEARS/1	CC-BY-SA-4.0
Steinmännchen / Antholzer See 03.01.2026	CC-BY-4.0

Query 2: creators (prefixes omitted)

```
SELECT ?creator (COUNT(DISTINCT ?fdo) AS ?models)
WHERE { ?fdo a fdo:3DDataFDO ;
dct:creator ?agent .
?agent schema:name ?creator . }
GROUP BY ?creator
ORDER BY DESC(?models) ?creator
```

Creator	Models
Thiery, Florian	5
Distel, Anne-Karoline	1
b-unicycling	1

People

Louise Tharandt

Florian Thiery

Behind the work

Project, institute and networks

FDOx-Squirrel · LEIZA · NFDI4Objects · Research Squirrel Engineers

References

- De Smedt, K., Koureas, D., & Wittenburg, P. (2020). FAIR Digital Objects for Science: From Data Pieces to Actionable Knowledge Units. *Publications*, 8(2), Article 2. https://doi.org/10.3390/publications8020021
- Thiery, F. (2026). FAIR Digital Object Squirrel Implementation (FDOSquirrel). *Squirrel Papers*, 8(2), §4. https://doi.org/10.5281/zenodo.18441544
- fdo-squirrel: https://github.com/FDOx-squirrel/fdo-squirrel
- fdo-3d-packager: https://github.com/FDOx-squirrel/fdo-3d-packager